Ayan Antik Khan

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Research Interests

My research focuses on the mechanistic interpretability of language models. I build AI agents that automate interpretability research and study how interpretability can improve LLM safety and alignment. I am also interested in reasoning and bias in LMs.

Education

George Mason University — Fairfax, VA
Ph.D. in Computer Science, GPA: 4.00/4.00 — Aug 2025 – Present
Advised by Prof. Ziyu Yao

Bangladesh University of Engineering and Technology (BUET) — Dhaka, Bangladesh
B.Sc. in Computer Science and Engineering, CGPA: 3.85/4.00 — Apr 2018 – May 2023
Advised by Dr. Rifat Shahriyar

Research Experience

Graduate Research & Teaching Assistant, George Mason University — Aug 2025 – Present

- Led a first-author project on using LM agents to explain circuits in language models. The paper was accepted to Findings of EMNLP 2026 and presented as a poster at NEMI 2026.
- Won the top white-box prize at Aletheia's Quest 2026 (AI lie detection).

Publications

Can Language Model Agents be Helpful Circuit Explainers in Mechanistic Interpretability?
Ayan Antik Khan, Harsh Kohli, Yuekun Yao, Huan Sun, Ziyu Yao
Findings of EMNLP 2026

An Empirical Study on the Characteristics of Bias upon Context Length Variation for Bangla
Jayanta Sadhu*, **Ayan Antik Khan***, Abhik Bhattacharjee, Rifat Shahriyar
Findings of ACL 2024

Multi-ToM: Evaluating Multilingual Theory of Mind Capabilities in Large Language Models
Jayanta Sadhu*, **Ayan Antik Khan***, Noshin Nawal, Sanju Basak, Abhik Bhattacharjee, Rifat Shahriyar
arXiv preprint, 2024
* Equal contribution.

Industry Experience

Software Development Engineer, IQVIA — Jun 2023 – Jul 2025

- Front-end developer on the Orchestrated Analytics suite, working on KPI Library and AI Assistant.
- Promoted to Software Development Engineer II in Apr 2025.

Technical Skills

Languages: Python, C++, C, Java, TypeScript, SQL, Bash

ML & LLMs: PyTorch, Hugging Face Transformers, scikit-learn, TensorFlow, Pandas, Weights & Biases

Software & Tools: Git, AWS SageMaker, PostgreSQL, Angular, React, Django

Selected Awards

Top White-Box Prize, Aletheia's Quest AI Lie Detection Challenge — 2026
Gold Impact Award, IQVIA — 2024
Finalist, Bangla Complex Named Entity Recognition Challenge — 2023
RISE Research Grant, BUET, for undergraduate thesis — 2022
Dean's List (6 terms) and **University Merit Scholarship**, BUET — 2018 – 2023